

TESTING COORDINATION & PROJECT READINESS PACKAGE

HealthConnect Clinic Experience Lab – Week 7

Project Name	HealthConnect Clinic Experience Lab
Sponsor	AnalystLab Africa (Experience Lab Programme)
Project Manager	Leonel Devinci Ebaha Djoki
Week	Week 7
Phase	Testing, Refinement & End-to-End Validation
Document Version	1.0

Part 1: Week 6 to Week 7 Transition

1.1 Main Week 6 Output

The Project Integration & Coordination Package was delivered, comprising the Workstream Integration Map, Updated Timeline, Integration/Dependency Register, Integration Issue Log, Decision Log, Updated Risk Register, Cross-Track Coordination Record, Project Status Update, and Week 7 Testing Coordination Plan.

1.2 Most Important Result

The integration with the Data Science track on the treatment of incomplete demographic fields (risk R7, dependency D1). Two fields were dropped, one was imputed, and the arbitrage was documented in the Decision Log (DEC-01 and DEC-02). Risk R7 moved to Partially Mitigated.

1.3 Main Issue Identified

R7 remained at Partially Mitigated status: the residual uncertainty was whether dropping two demographic fields had caused a loss of predictive signal in the model. This could only be verified through actual model testing.

1.4 Component Requiring Testing

The candidate model produced by Data Science, specifically the impact of the feature engineering decisions taken jointly with the PM in Week 6.

1.5 Track Relevant to Week 7

Data Science remains the primary testing partner. Data Analytics, ML Engineering and Generative AI are also tested, with results consolidated by the PM.

1.6 What Will Be Tested

Whether the model performs adequately with the adjusted feature set, whether the pipeline processes data correctly end to end, whether the dashboard and KPIs are accurate, and whether the AI assistant remains compliant and consistent.

1.7 What Will Be Improved or Validated

The status of R7 (from Partially Mitigated to a definitive assessment), the status of all remaining open risks (R1 through R8), and the overall project readiness for the Week 8 final presentation.

Testing Readiness

A. Component Identification

The primary component tested by the PM track is the cross-track integration between Data Analytics, Data Science and ML Engineering, as coordinated through the Workstream Integration Map. The intended purpose of this integration is to ensure that analytical findings translate into sound modelling decisions and that the resulting model can be processed reliably through the ML pipeline. The secondary testing focus covers the Generative AI assistant's compliance with the Knowledge Base.

B. Testing Objective

Determine whether the decisions made during the Week 6 integration phase (specifically DEC-01 and DEC-02) produced a model that performs adequately for the HealthConnect use case, and whether the four tracks' outputs are collectively ready for the Week 8 final presentation.

C. Consolidated Testing Results

Track	Test Performed	Expected Result	Actual Result	Result	Action / Notes
Data Analytics	Validate KPI calculations against the dataset and verify dashboard accuracy	KPIs match underlying data; dashboard filters return correct values	All three KPIs verified; one dashboard filter (distance bracket) returned a minor rounding discrepancy, corrected	Pass	Rounding issue fixed; dashboard retested and confirmed accurate
Data Science	Test candidate model on test set; assess impact of Week 6 feature engineering (R7)	Stable or improved performance compared to Week 5 baseline, despite removal of two features	Performance globally stable; slight recall decrease on patients with no prior appointment history	Pass (with reservation)	Recall limitation documented as accepted trade-off (DEC-05); not blocking for HealthConnect use case
ML Engineering	End-to-end pipeline test:	Pipeline processes	Pipeline ran successfully;	Pass	No issue identified;

	raw data to preprocessing to prediction to output	input correctly and produces expected output format	output format and values consistent with direct model testing		pipeline ready for Week 8
Generative AI	Test response consistency, Knowledge Base compliance, escalation behaviour, and unsupported requests	Responses accurate and within scope; escalation triggers correctly; unsupported requests handled appropriately	One prompt produced a slightly ambiguous response on a rescheduling edge case; all other tests passed	Pass	Prompt refined for the rescheduling scenario; retested and confirmed clear

D. Testing Evidence

- Updated Risk Register with definitive statuses for all nine risks.
- Updated Dependency Register showing all four dependencies as tested.
- Integration and Issue Log with six entries, including resolution status.
- Decision Log with three new entries (DEC-05 through DEC-07).
- Testing Activity Tracker provided as a separate Excel file.

1. Week 7 Testing and Validation Plan

The testing plan follows the approach outlined in the Week 7 Testing Coordination Plan prepared at the end of Week 6. Each of the four technical tracks was asked to test its primary Week 6 output and report results to the PM for consolidation. The PM's role was to coordinate the testing activities, track issues, facilitate cross-track resolution where needed, and produce the consolidated readiness assessment.

Testing Approach by Track

- Data Analytics: validate KPI calculations against the source dataset; verify dashboard accuracy and filter behaviour.
- Data Science: test the candidate model on the held-out test set; evaluate classification metrics; assess the impact of the Week 6 feature engineering decisions on model performance.
- ML Engineering: run the integrated pipeline end to end (raw data through preprocessing through prediction through output); verify output format and consistency with direct model testing.
- Generative AI: test response consistency across similar questions; verify Knowledge Base compliance; test escalation triggers and unsupported request handling.
- Project Management: consolidate all test results; update the risk and issue registers; assess project readiness for Week 8.

2. Updated Timeline

The Updated Timeline is provided as a separate Excel file (W7_Updated_Timeline_HealthConnect.xlsx). Weeks 4 through 6 are marked as Complete. Week 7 is marked as In Progress. Week 8 (Final Integration and Presentation) is Planned. The calendar remains confirmed as established in Week 6.

3. Testing Activity Tracker

The detailed Testing Activity Tracker is provided as a separate Excel file (W7_Testing_Activity_Tracker_HealthConnect.xlsx). It extends the Task Tracker structure used in previous weeks with a dedicated testing sheet, recording each test scenario, expected and actual results, pass/fail status, issues identified, actions taken, and retest outcomes across all four technical tracks.

4. Integration and Issue Log

ID	Description	Raised	Status	Resolved	Resolution
ISS-01	Feature loss due to Week 6 data quality trade-off	Week 5	Closed	Week 7	Model tested; stable performance confirmed with a documented recall limitation on one segment
ISS-02	No official programme calendar	Week 4	Closed	Week 6	Calendar confirmed by the Week 6 brief
ISS-03	Tight testing timeline if Week 6 overran	Week 6	Closed	Week 7	Testing completed within Week 7 as planned
ISS-04	Recall limitation on patients with no prior appointment history	Week 7	Open (accepted)	n/a	Accepted as a known limitation; not blocking; to be disclosed in Week 8 presentation
ISS-05	Minor dashboard rounding discrepancy on distance	Week 7	Closed	Week 7	Corrected by Data Analytics; retested and confirmed

	bracket filter				accurate
ISS-06	Ambiguous GenAI response on rescheduling edge case	Week 7	Closed	Week 7	Prompt refined by Generative AI; retested and confirmed clear

Summary: six issues tracked across the project lifecycle. Four closed (ISS-01, ISS-02, ISS-03, ISS-05, ISS-06 resolved through testing or calendar confirmation). One open and accepted (ISS-04, recall limitation). No blocking issue remains.

5. Updated Risk and Dependency Register

5.1 Risk Register

ID	Risk	Prob.	Impact	Score	Response	Mitigation	Status
R1	Upstream track delay blocking downstream	Medium (2)	High (3)	6	Mitigate	Regular sync points; early alerts	Closed (no delay materialized through Week 7)
R2	Inconsistency between track deliverables	Medium (2)	Medium (2)	4	Mitigate	Weekly consolidation by PM	Closed (testing confirmed consistency)
R3	GenAI scope drift outside Knowledge Base	Low (1)	High (3)	3	Mitigate	Scope check at each follow-up	Closed (testing confirmed compliance)
R4	Calendar differs from 1 phase/week	n/a	n/a	n/a	n/a	Confirmed by Week 6 brief	Closed (Week 6)
R5	PM overloaded coordinating	Low (1)	Medium (2)	2	Mitigate	Prioritization via stakeholder matrix	Closed (managed through 4)

	four tracks						weeks)
R6	Unplanned technical integration	Low (1)	Medium (2)	2	Accept	Scope review during integration	Closed (integration phase completed, only R7 materialized)
R7	Data quality limits ML features	Medium (2)	Medium (2)	4	Mitigate	PM business input + DS feature adjustment	Validated with documented limitation
R8	Tight testing window	Medium (2)	Medium (2)	4	Mitigate	PM confirmed readiness with all tracks	Closed (testing completed on time)
R9	Recall limitation on patients with no prior history	Low (1)	Low (1)	1	Accept	Documented as known limitation; to mention in Week 8 presentation	NEW (Week 7), accepted

Summary of changes from Week 6: R1, R2, R3, R5 closed based on testing evidence (no delay, no inconsistency, no scope drift, no overload materialized). R6 closed (integration phase concluded without incident beyond R7). R7 moved from Partially Mitigated to Validated with documented limitation. R8 closed (testing completed on time). R9 added as a new, accepted, low-severity limitation.

5.2 Dependency Register

ID	Dependency	Type	Week 6 Status	Week 7 Status	Notes
D1	Data Science needs Data Analytics outputs for	Sequential	Integrated (residual limitation)	Integrated and tested	Model tested with adjusted features; performance

	features				validated
D2	ML Engineering needs ML problem from Data Science	Sequential	Monitored	Integrated and tested	Pipeline tested end-to-end with candidate model; output validated
D3	Generative AI depends on Knowledge Base only	External	Confirmed independent	Tested and confirmed	Compliance verified through structured testing
D4	Programme calendar	External	Resolved	Resolved	Calendar confirmed since Week 6

All four dependencies have reached their final tested status. D1 and D2 are now fully integrated and tested. D3 is confirmed through structured testing. D4 was resolved in Week 6.

6. Issue Resolution Status

- ISS-01 (feature loss): closed. Model testing confirmed stable performance with a narrow, documented limitation.
- ISS-02 (calendar): closed since Week 6.
- ISS-03 (tight timeline): closed. Testing completed within the planned window.
- ISS-04 (recall limitation): open, accepted. Narrow scope (one patient segment), not blocking. To be disclosed in the Week 8 presentation.
- ISS-05 (dashboard rounding): closed. Corrected and retested within Week 7.
- ISS-06 (GenAI ambiguous response): closed. Prompt refined and retested within Week 7.

7. Decision Log

ID	Date	Decision	Rationale	Made By	Impact
DEC-05	Week 7	Accept the recall limitation on patients with no prior history as a documented trade-off rather than reopening feature engineering	Reopening the feature set would delay the project past the Week 8 deadline; the limitation is narrow (one segment) and the overall model performance	PM + DS	R9 created; ISS-04 logged

			remains stable		
DEC-06	Week 7	Close risks R1, R2, R3, R5, R6, R8 based on testing evidence	All four tracks completed their testing within the planned timeline; no upstream delay, no cross-track inconsistency, no scope drift, no overload, no unplanned integration, and no testing overrun materialized	PM	Risk register simplified for Week 8
DEC-07	Week 7	Declare the project ready for Week 8 final integration and presentation	All tracks have completed testing; all critical issues resolved; one accepted limitation documented (R9/ISS-04); no blocking item remaining	PM	Formal readiness assessment produced

8. Cross-Track Coordination Record

Data Analytics

The Data Analytics track completed its validation of the three KPIs and the dashboard. One minor rounding discrepancy was identified in the distance bracket filter, corrected, and retested successfully (ISS-05). No other issue was found. The validated KPI values were cross-checked by the PM against the model testing results from Data Science to confirm internal consistency across the two tracks.

Data Science

This was the most significant testing exchange of the week. The Data Science intern tested the candidate model on the held-out test set and shared the evaluation metrics with the PM. Overall model

performance was stable compared to the Week 5 baseline, confirming that the feature engineering decisions of Week 6 (DEC-01, DEC-02) did not degrade the model in a meaningful way. However, a slight decrease in recall was observed specifically for patients with no prior appointment history. The PM and Data Science jointly assessed this finding and decided to accept it as a documented trade-off rather than reopening the feature engineering cycle (DEC-05). This decision moved R7 to its final status (Validated with documented limitation) and created a new low-severity risk (R9) and issue (ISS-04).

ML Engineering

The ML Engineering track ran the integrated pipeline end to end and confirmed that data flowed correctly from raw input through preprocessing through prediction through output. The output format and values were consistent with the results obtained by Data Science during direct model testing. No integration issue was identified. Dependency D2 moved to Integrated and tested.

Generative AI

The Generative AI track completed its structured testing. One edge case was identified: an ambiguous response on a rescheduling scenario that could confuse a patient. The prompt was refined and the scenario was retested successfully (ISS-06). Knowledge Base compliance was verified across all test scenarios. Dependency D3 was confirmed through testing.

9. Testing Progress and Status Report

Overall Status

The HealthConnect Clinic Experience Lab has completed the Testing, Refinement and End-to-End Validation phase. All four technical tracks have tested their primary outputs, resolved identified issues (where applicable), and confirmed readiness for the Week 8 final presentation.

Testing Summary

- Tests performed: 4 (one per technical track, coordinated by the PM).
- Results: 3 Pass, 1 Pass with reservation (Data Science).
- Issues identified: 3 new (ISS-04, ISS-05, ISS-06).
- Issues resolved within Week 7: 2 (ISS-05, ISS-06).
- Issues accepted: 1 (ISS-04, recall limitation).
- Risks closed in Week 7: 6 (R1, R2, R3, R5, R6, R8).
- Risks validated: 1 (R7).
- Risks added: 1 (R9, low severity, accepted).

Schedule Adherence

The project remains fully aligned with the confirmed calendar. All Week 7 activities were completed within the planned timeframe. No schedule variance has been observed across the entire project lifecycle (Weeks 4 through 7).

10. Week 8 Readiness Assessment

The following assessment evaluates whether the HealthConnect project is ready to move into its final phase: integration and presentation.

Readiness by Track

- Data Analytics: ready. KPIs validated, dashboard accurate, findings consistent with the modelling results.
- Data Science: ready (with one documented limitation). Model tested, performance stable, limitation on one segment acknowledged and documented.
- ML Engineering: ready. Pipeline tested end to end, output validated, no integration issue remaining.
- Generative AI: ready. Assistant tested, prompt refined, Knowledge Base compliance confirmed, escalation behaviour verified.

Readiness by Governance Criteria

- Risk register: all critical and medium risks closed or validated. One low-severity accepted risk (R9).
- Dependency register: all four dependencies at their final tested status.
- Issue log: no blocking issue. One accepted limitation (ISS-04) to disclose during the presentation.
- Decision log: seven decisions documented across the project, providing a clear audit trail.

Overall Assessment

The project is ready for Week 8 final integration and presentation. All tracks have completed their development, integration and testing cycles. The one remaining limitation (recall on a specific patient segment) is documented, understood, and accepted rather than hidden. This is a strength, not a gap: it demonstrates that the project team identified what works and what does not, rather than presenting an artificially clean result.

11. Outstanding Actions Before Final Integration

- Each track to prepare its contribution to the final HealthConnect presentation (slides, key findings, demo where applicable).
- PM to prepare the presentation coordination framework and the consolidated project narrative.
- PM to prepare the final project closeout summary.
- ISS-04 (recall limitation) to be explicitly mentioned in the final presentation as a known limitation.

Mandatory HC-POD Cross-Track Testing and Refinement

- Track collaborated with: Data Science.
- Project dependency: D1 (Data Analytics to Data Science), R7.
- Component tested: the candidate model, specifically the impact of the Week 6 feature engineering decisions (DEC-01, DEC-02) on model performance.
- Information received: Data Science's model evaluation metrics on the test set, including per-segment performance breakdown.
- Information provided: PM's assessment of the business acceptability of the observed recall limitation, based on HealthConnect's operational priorities.
- Testing activity completed: model testing on the test set, followed by joint analysis of the recall limitation, followed by a shared decision to accept the trade-off.
- Issue identified: slight recall decrease on patients with no prior appointment history (ISS-04).
- Action taken: documented as an accepted trade-off (DEC-05) rather than reopening feature engineering.
- Retest result: not required; the limitation is narrow and accepted, not a defect requiring correction.
- What changed: R7 moved from Partially Mitigated to Validated with documented limitation; D1 moved to Integrated and tested; R9 created as a new accepted risk.
- Evidence: Updated Risk Register, Decision Log (DEC-05), Issue Log (ISS-04), Testing Activity Tracker.
- How this improved HealthConnect: the project now has a tested, honest assessment of its prediction model rather than an untested assumption that everything works perfectly.